18/5/2022 (E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: January 2022 – May 2022 Examination: ESE Examination	Duration: 3hrs
Programme code: 01&04 Programme: B Tech Comp/IT	Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: COMP/IT	
Course Code: 116U01C401 /116U04C401	Name of the Course: Probability, Statistics and Optimization Techniques	
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary		

Q No			MAX MARKS
Q1	a	If X_1 has mean 5 and variance 5, X_2 has mean -2 and variance 3, find $E(2X_1+3X_2-5)$, $V(2X_1+3X_2-5)$	5
	b	**Solve any THREE of the following**	21
	(i)	Determine the constant 'a' and find mean, $P(4 \le x \le 7)$ if the distribution function of a continuous random variable is defined as $f(x) = \frac{a}{x^5}$, $2 \le x \le 10$	
	(ii)	If the height of 1000 students is normally distributed with mean 69 inches and standard deviation 4 inches. Find the expected number of students having heights: i) greater than 67 inches, ii) less than 68 inches, iii) between 65 & 71 inches	
	(iii)	The number of phone calls coming in to a telephone exchange between 2 & 4 P.M. say X is a random variable has Poisson distribution with parameter 2. Similarly the number of phone calls coming between 4 & 6 P.M. say Y is a random variable has Poisson distribution with parameter 6. If X& Y are independent Poisson random variables find the probability that during 2&6 P.M. there will be i) no phone calls at all ii) more than 3 calls. (iii) at most two calls	
	(iv)	A box to be constructed so that its height is 12 inches and its base is X inches by X inches. If X has a uniform distribution over the interval (2, 10), then what is the expected volume of the box in cubic inches?	
	(v)	The joint probability distribution function of (X,Y) is given by $f(x,y) = e^{-(x+y)}$ $0 \le x$, $0 \le y$ Compute$P(X > 2), P(1 < X + Y < 3))$	

Q2 a A data for selection of students regarding placement is given below. Find the probability that a boy is selected for the placement and log of odds of this probability **(5)**

Students	Selection in placement		Total
	yes	no	
Girls	753	102	855
Boys	382	158	540
Total	1145	250	135

b **Solve any ONE of the following** **(7)**

(i) Calculate the correlation coefficient from the following data.

x	23	27	28	29	30	31	33	35	36	39
y	18	22	23	24	25	26	28	29	30	32

(ii) Obtain two lines of regression and coefficient of correlation from the following data-

X	65	66	67	68	69	70	72	67
y	67	68	65	72	72	69	71	66

Q3 a Two samples are drawn from two different population gave the following results. Find 95% confidence limits for the difference between the population means. **(5)**

	Size	Mean	S.D
Sample I	400	124	14
Sample II	250	120	12

b **Solve any TWO of the following** **(14)**

(i) Intelligence tests of two groups of boys & girls obtained from two normal populations having the same standard deviations gave the following results. Test at 1% level of significance whether the boys perform better than the girls.

	Size	Mean	S.D
Girls	121	84	10
Boys	181	81	12

(ii) A certain injection administered to 12 patients resulted in the following change of blood pressure 5,2,8, -1,3,0,6, -2,1,5,0,4. Can be concluded that the injection will be in general accompanied by an increase in blood pressure at 5% LOS?

(iii) From the following table, showing the number of plants having certain character, test the hypothesis that the flower colour is independent of flatness of leaf.

	Flat leaves	Curved leaves	Total
White Flowers	99	36	135
Red Flowers	20	5	25
Total	119	41	160

Q4	a	Construct the Dual of the following LPP Maximize $z = 5x_1 + 2x_2 - 3x_3$ Subject to $2x_1 - 2x_2 + x_3 \geq 4 \quad 2x_1 + x_3 \leq 8 \quad x_1 + x_2 + 3x_3 = 20$ $x_1, x_3 \geq 0, \quad x_2$ unrestricted	5
	b	**Solve any THREE of the following**	21
	(i)	Using Simplex method solve the following LPP Maximize $z = 3x_1 + 2x_2 + 5x_3$ Subject to $x_1 + x_2 + x_3 \leq 9 \quad 2x_1 + 3x_2 + 5x_3 \leq 30 \quad 2x_1 - x_2 - x_3 \leq 8$ $x_1, x_2, x_3 \geq 0$	
	(ii)	Using Big M method solve the following LPP Maximize $z = 6x_1 + 4x_2$ Subject to $2x_1 + 3x_2 \leq 30, \ 3x_1 + 2x_2 \leq 24, x_1 + x_2 \geq 3, \ x_1, x_2 \geq 0$	
	(iii)	Using Duality Solve the following linear programming problem Minimize $z = 4x_1 + 3x_2 + 6x_3$ Subject to $x_1 + x_3 \geq 2, \quad x_2 + x_3 \geq 5, \quad x_1, x_2, x_3 \geq 0$	
	(iv)	Using Dual simplex method Solve the following linear programming problem Minimize $z = 2x_1 + 2x_2 + 4x_3$ Subject to $2x_1 + 3x_2 + 5x_3 \geq 2, \quad 3x_1 + x_2 + 7x_3 \leq 3, \quad x_1 + 4x_2 + 6x_3 \leq 5$ $x_1, x_2, x_3 \geq 0$	
	(v)	Solve the following NLPP Maximize $z = 2{x_1}^2 - 7{x_2}^2 + 12x_1x_2$ Subject to $2x_1 + 5x_2 \leq 98, \ x_1, x_2 \geq 0$	
Q5	a	In a bank cheques are cashed at a single 'teller' counter. Customers arrive at the counter in a Poisson manner at an average rate of 30 customers per hour. The teller takes, on an average, a minute and a half to cash a cheque The service time has been shown to be exponentially distributed. Calculate the % of time the teller is busy.	3

	b	**Solve any TWO of the following**	14
	(i)	Patients arrive at a clinic according to Poissondistribution at the rate of 30 patients per hour. The waiting room does not accommodate more than 14 patients. The examination time per patient is exponential with mean rate of 20 per hour. (a) Find number of patients in the clinic before the examination (b) What is the probability that an arriving patient will not wait. (c) What is the expected waiting time until a patient is discharged from the clinic ?	
	(ii)	Trucks arrival at a factory is for collecting finished goods that are supposed to be transported to distant markets. As and when they come they are required to join awaiting line and are served on first come, first served basis. Trucks arrive at the rate of 10 per hour where as the loading rate is 15 per hour. It is also given that arrivals are Poisson and loading is exponentially distributed. (a) Transporters have complained that their trucks have to wait for nearly 12 minutes at the plant. Examine whether the complaint is justified. (b) Determine the number of trucks waiting in the queue before getting loaded. (c) Find the probability that a truck cannot be loaded immediately.	
	(iii)	Customer arrives at a box office window, being manned by a single individual, according to a Poisson input process with a mean rate of 30 per hour. The time required to serve a customer has an exponential distribution with a mean of 90 seconds Find the average time spent by a customer. Also determine the average number of customers in the system and the average queue length	

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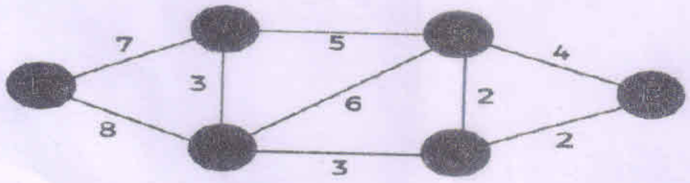
23|05|2022 (E)

Maximum Marks: 100	Semester: January 2022 – May 2022 Examination: ESE Examination		Duration:3 Hours
Programme code: 04 Programme: B Tech Information Technology		Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Information Technology	
Course Code: 116U04C403	Name of the Course: Analysis of Algorithms		
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary			

Question No.		Max. Marks
Q1 (a)	Discuss the analysis of the algorithm using Big – O asymptotic notation. Give logarithmic example of run time analysis.	10
Q1 (b)	Solve recurrence relation $T(n) = 2T(n/2)+n^2$.	10
Q2 (a)	Explain selection sort algorithm. Discuss the time complexity of selection sort in best case, average case and worst case scenario.	10
Q2 (b)	Explain Divide and conquer technique. Explain merge sort algorithm with example.	10
	Or	
	Prove that the average case complexity of quick sort for sorting n elements is "n log n"	10
Q3 (a)	What is Minimum Cost Spanning Tree? Calculate minimum spanning tree of the below graph using kruskal's algorithm. 	10
Q3 (b)	Given two sequences X [1...m] and Y [1.....n]. Find the longest common subsequences to both. x: A B C B D A B y: B D C A B A	10
	Or	
	Differentiate between dynamic programming and greedy method. Explain Travelling sales man problem in Dynamic programming.	10

Contd...

Q4 (a)	Explain sum of subset problem. Solve the sum of subset problems using backtracking algorithmic strategy for the following data: n = 4 W =(w1, w2, w3, w4) = (11, 13, 24, 7) and M = 31.	10
Q4 (b)	Explain Hamiltonian Circuit Problem with example	10
Q5	Write short note:(**any two**) 1)N queens Problem 2)NP reducibility 3)NP completeness 4)Strassen's matrix multiplication	20

Q4 (a)	Explain sum of subset problem. Solve the sum of subset problems using backtracking algorithmic strategy for the following data: n = 4 W =(w1, w2, w3, w4) = (11, 13, 24, 7) and M = 31.	10
Q4 (b)	Explain Hamiltonian Circuit Problem with example	10
Q5	Write short note:(**any two**) 1)N queens Problem 2)NP reducibility 3)NP completeness 4)Strassen's matrix multiplication	20

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Maximum Marks: 100	Semester: January 2022 – May 2022 Examination: ESE Examination		Duration: 3 Hr.
Programme code: 04 Programme: Information Technology		Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: IT	
Course Code: 116U04C404	Name of the Course: Advanced Databases		
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary			

Question No.		Max. Marks
Q1 (a)	Explain the various I/O parallelisms.	10
Q1 (b)	Consider a relation that fragmented horizontally by Plant_number: employee (name, address, salary, plant_number) Assume that each fragment has two replicas: one stored at the Mumbai site and one stored locally at the plant site. Write the processing strategy for following queries entered at the New Delhi. i. Find all employees at the Chennai. ii. Find the lowest-paid employee in the company. **OR** Consider the following details. Employee table is at site 1 with 10,000 rows. Each row size is 100 bytes. table size is 10^6 bytes. Department table is at site 2 with 100 rows. Each row size is 35 bytes. Table size is 3500 bytes. i. Find optimum solution for data transfer if following query is executed from site 3. **Query:** For each employee retrieve the emp_name and dept_name where employee works. size of result tuple is 40 bytes. ii. Find the cost of data transfer over the network.	10
Q2 (a)	Briefly discuss about how to design an object relational database?	10
Q2 (b)	Explain **any one** from the following i. NoSQL ii. Spatial Database	10
Q3 (a)	Explain star schema. Draw Star Schema for supermarket. **OR** Explain snowflack schema. Discuss snowflack schema for hospital.	10

Q3 (b)	Consider Type 1, 2, 3 changes to update the following dimension table. Customer (Customer_id, Name, Marital status, Address, mobile number). i. Previous name: Smith Judy. Change the previous name to Smith John. ii. Change the Martial status of the above customer from single to Married from the date 7th august 2018. iii. Change address of the above customer from Chicago to New Delhi from 2nd Dec 2018.	10
Q4 (a)	Explain in detail issues in data cleaning.	10
Q4 (b)	Explain the different techniques for data loading with suitable example.	10
Q5 (a)	Write the most relevant differences between i. OLTP Vs OLAP ii. Data warehouse modeling Vs Operational database modeling	10
Q5 (b)	Consider a Data warehouse for Library where there are three dimensions. 1. Student, 2. Books, 3.Publication and two measures 1. Date of issue book and 2. Fine (if book not return in stipulated period). For this example describe the following OLAP operations with suitable diagram, i. Rollup ii. Drilldown iii. Dice iv. Slice v. Pivot **OR** Consider Apple company products, for which dimensions [Product, Customer, Day]. Which OLAP operations should be performed to execute following queries, also visualize the result of queries. (Assume suitable data) i. Display the total sale of all products for past five years in all stores ii. Compare total sales amount of Laptop and Mobile for 2020 to 2021.	10

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Maximum Marks: 100	Semester: January 2022 – May 2022 Examination: ESE Examination	Duration: 3 Hrs
Programme code: 46 Programme: Minor in IT	Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: IT	
Course Code: 116m46C401	Name of the Course: Building Blocks of Information Technology	
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary		

Question No.		Max. Marks
Q1 (a)	Explain the algorithm of binary search technique with example	10
Q1 (b)	Explain Quick Sort algorithm and give a suitable example	10
Q2 (a)	Explain ISO-OSI Model with neat diagram	10
Q2 (b)	Describe about the services provided by data link layer **OR** Differentiate the following: a) TCP and UDP b) Preemptive and Non-preemptive scheduling	10
Q3 (a)	Explain external fragmentation with suitable example. What are the causes for external fragmentation? **OR** What is a scheduler? Describe different CPU scheduler.	10
Q3 (b)	Distinguish between (any two) a) Process and Program b) Multiprogramming and multiprocessing c) Job scheduling and CPU scheduling.	10
Q4 (a)	Describe the LRU page replacement algorithm, assuming there are 3 frames and the page reference string is 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1. Find the number of page faults.	10
Q4 (b)	Explain the concept of Class Diagram with the help of following points - Classes, Associations, Attributes, Operations and Generalizations. **OR** Explain System Testing and Unit Testing	10
Q5 (a)	Describe following methods of Testing - a) White Box Testing b) Black Box Testing	10
Q5 (b)	What is Design Pattern? Explain Creational and Structural Design Pattern in detail **OR** Explain software design architecture design process	10

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Maximum Marks: 100	Semester: January 2022 – May 2022 Examination: ESE Examination		Duration: 3 Hrs.
Programme code: 42 Programme: Minor Programme in Augmented and Virtual Reality		Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP/ETRX/EXTC/IT/MECH	
Course Code: 116m42C401	Name of the Course: Sensors and Actuators in Augmented and Virtual Reality		
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary			

Question No.		Max. Marks
Q1 (a)	Explain working of IMU in detail with Interfacing diagram	10
Q1 (b)	Write a short note on 360 degree cameras **OR** Explain working of stereo Camera in detail	10
Q2 (a)	Explain concept of VR headset with one suitable example	10
Q2 (b)	Explain any 3 sensors used in Virtual reality **OR** Explain any 3 actuators in Virtual reality	10
Q3 (a)	Compare CMOS and CCD sensor **OR** Explain any one hardware interfacing of sensor with Raspberry Pi	10
Q3 (b)	Explain any one head tracking Mechanism	10
Q4 (a)	Explain construction of VR gloves with flex sensors	10
Q4 (b)	Explain softwares used for interfacing of Sensors and actuators with VR applications	10
Q5 (a)	Explain CAVE (Cave Automated Virtual Environment)	10
Q5 (b)	Write short note on any **two out of four** I. Unity II. Unreal III. Sketup IV. Arduino	10

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Maximum Marks: 100	Semester: January 2022 – May 2022 Examination: ESE Examination	Duration: 3 Hours
Programme code: 66 Programme: Honours in Artificial Intelligence	Class: SY	Semester: IV (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: IT
Course Code: 116h66C401	Name of the Course: Fundamentals of Data Science	
Instructions: 1)Draw neat diagrams 2)Assume suitable data if necessary		

Question No.		Max. Marks
Q1 (a)	Discuss the nature of the time series data and spatial data. Give valid example of each. OR What is the nominal data; give examples considering any 5 attributes (with its values) which holds the nominal data.	10
Q1 (b)	Explain any three transformations required to make the data suitable for the data analytics / mining?	10
Q2 (a)	What is the purpose of finding the similarity or dissimilarity among the data elements? Explain the similarity measures applicable for binary data with suitable example.	10
Q2 (b)	For the below given data points pairs, calculate the Euclidean and Manhattan distance, i) A(0, 4), B(6, 2) ii) B(6, 2), C(9, 1) OR Find edit distance between the following pair of string, consider two operations, Insert and Delete: i) "abcdef" and "bcdesg" ii) "AABBCCDD" and "ABCDD"	10
Q3 (a)	Use the two methods below to normalize the following group of data: 200,300,400,600,1000 (i) Min-max normalization by setting min=0 and max=1 (ii) Z-score normalization.	10
Q3 (b)	Discuss any four applications of Histogram with suitable example. OR Why to discretize the data? Discuss any three methods for data discretization with example.	10

Q4 (a)	Why there is a need of data reduction in data analytics? Discuss any three techniques for the data reductions with suitable example.	10
Q4 (b)	What is an outlier? With suitable example demonstrate any one of the outlier detection method.	10
Q5 (a)	How does the icon based visualization technique works? Give any two applications where this visualization technique is most suitable.	10
Q5 (b)	Discuss in detail (Any Two). i. Dashboard ii Scoreboard iii Frequency polygon iv. Chart Vs Graph	10